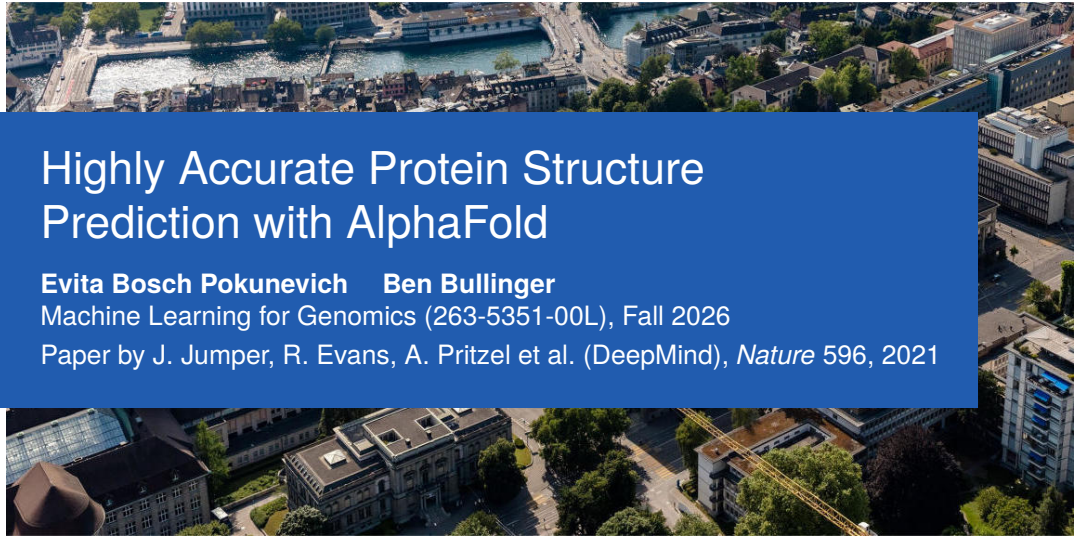
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Highly Accurate Protein Structure Prediction with AlphaFold

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Machine Learning for Genomics (263-5351-00L), Fall 2026

Paper by J. Jumper, R. Evans, A. Pritzel et al. (DeepMind), *Nature* 596, 2021

Outline

1. The problem
2. The network
3. Training
4. What makes it work
5. Conclusion

Sequence in, structure out

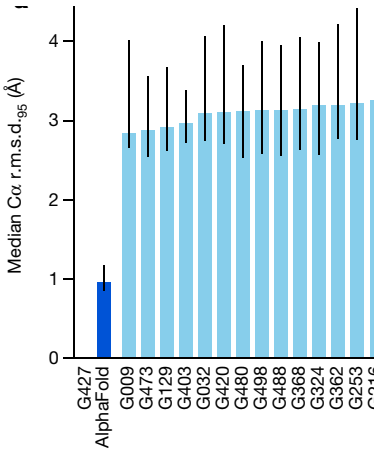
MTYKLILNGK… $\longmapsto$ all-atom 3D coordinates

A structure takes months of experimental effort; sequences number in the billions.

Fifty years of the protein folding problem, scored blind at CASP.

J. Jumper et al., *Nature* 596, 2021. C. B. Anfinsen, 1973.

CASP14: atomic accuracy, alone

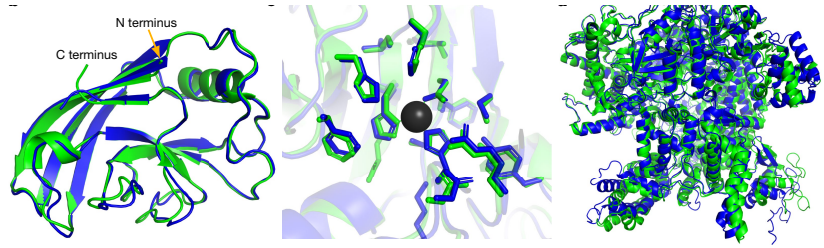


Median backbone accuracy 0.96 Å r.m.s.d.$_{95}$; the next best method 2.8 Å.

A carbon atom is about 1.4 Å wide.

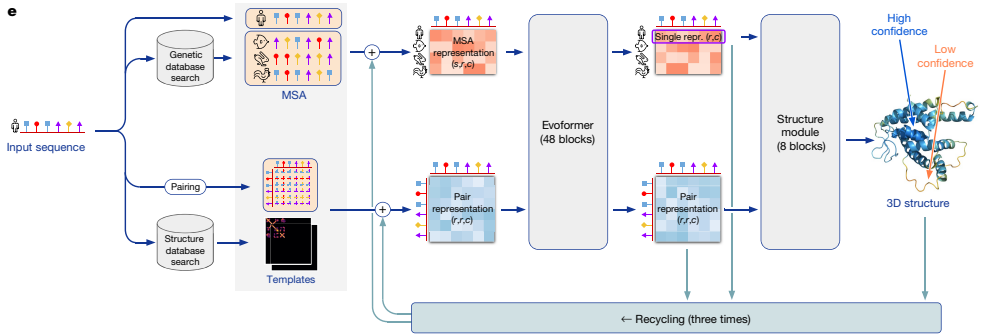
Error at the scale of the atoms themselves.

Predictions on top of experiment



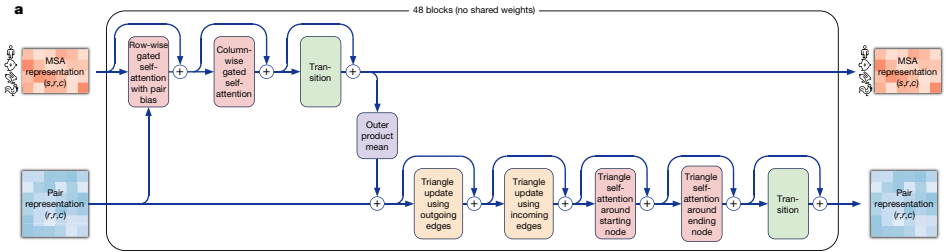
Backbone, side chains, and a 2,180-residue chain, without intervention.

One pipeline, end to end



MSA and templates in, coordinates out; the whole stack recycled three times.

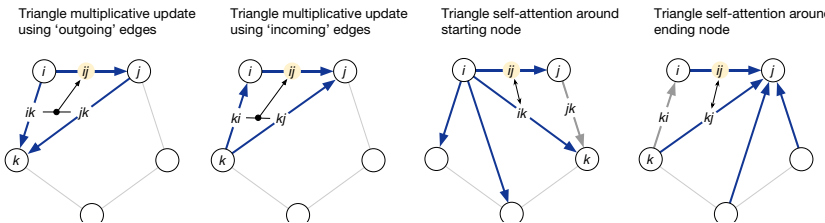
The Evoformer block



An MSA representation and a pair representation, exchanging information.

48 blocks, no shared weights.

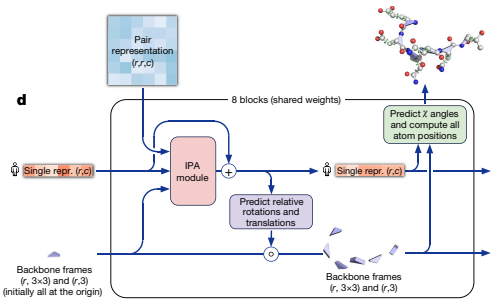
Pairs updated through triangles



Every edge ij is updated through the third edges ik, kj.

The triangle inequality, built into attention.

The structure module: residues as rigid bodies



Each residue is a rotation and a translation — a residue gas, refined all at once.

Invariant point attention keeps the update equivariant.

No chain constraint during refinement.

FAPE: a loss over local frames

$$\text{FAPE} = \text{mean}_{k,i} \left\| (R_k, \mathbf{t}_k)^{-1} \mathbf{x}_i - (R_k^{\text{true}}, \mathbf{t}_k^{\text{true}})^{-1} \mathbf{x}_i^{\text{true}} \right\|$$

Every atom scored in the frame of every residue, clamped L^1.

Invariant to pose, and the source of AlphaFold's chirality.

Trained beyond the PDB

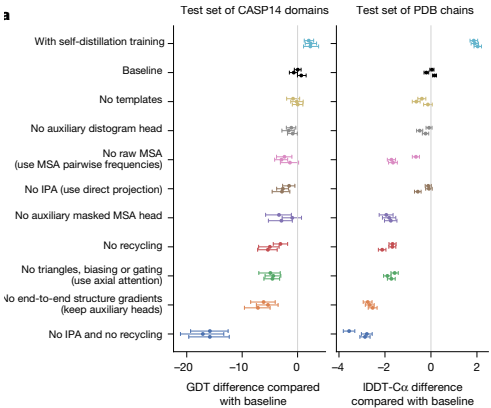
Self-distillation: predict 350,000 Uniclust structures, keep the confident ones, retrain.

A BERT-style masked-MSA loss, learned jointly, in place of hand-crafted covariation statistics.

The unlabelled sequence universe becomes training data.

Noisy-student self-distillation: Q. Xie et al., 2020. BERT: J. Devlin et al., 2019.

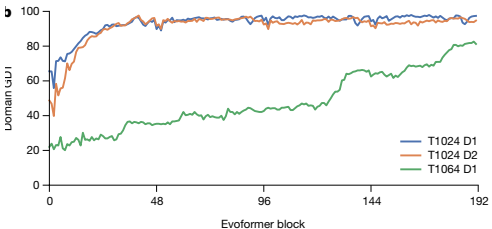
Every piece earns its accuracy



Ablating self-distillation, IPA, recycling, or the triangle updates each costs accuracy.

No single trick; a stack of compounding ones.

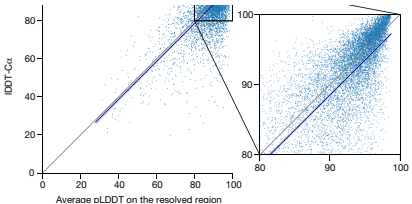
The structure forms inside the trunk



A frozen structure module read out at every block: most targets fold in the first few.

ORF8 of SARS-CoV-2 needs all 48.

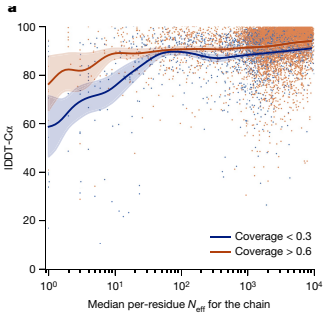
The model knows when it is right



pLDDT, predicted per residue, tracks the true lDDT-Cα.

Calibrated confidence made the proteome-scale database usable.

Where it still fails



Accuracy drops below about 30 effective sequences in the MSA.

Weak on chains shaped by hetero-contacts.

The evolutionary record is still an input, not an option.

Summary

Geometric and evolutionary inductive bias, learned end to end, reaches atomic accuracy.

Triangles for the pair geometry; frames and FAPE for the atoms; recycling throughout.

A GPU minute per structure, with calibrated confidence.

Open then, answered since: complexes (AlphaFold-Multimer), ligands and nucleic acids (AlphaFold 3).

J. Jumper et al., “Highly accurate protein structure prediction with AlphaFold,” *Nature* 596, 583–589 (2021).